

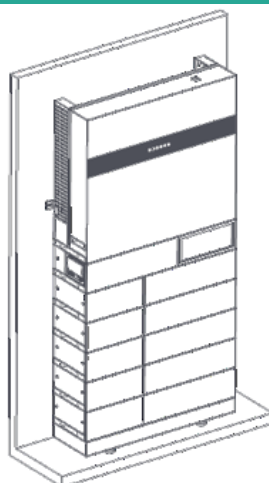
Alfred-10

Capacity

9.6-14.4 kWh

16.8-24 kWh

STRONG

ENERGY


PV-Input

Max. PV power (recommended)	15 kWp
Max. PV input voltage	1000 V
PV startup voltage	150 V
MPPT voltage range	160-950 V
No. of MPPTs	2
Max. no. of PV-strings per MPPT	1 + 2
Max. input current per MPPT	20 A + 30A
Max. short circuit current per MPPT	30 A + 40A

Battery

Cell chemistry	LFP (Lithium Iron Phosphate)	
No. of battery modules	4 - 6	7 - 10
Nominal capacity	9.6-14.4 kWh	16.8-24 kWh
Nominal voltage	200-300 V	350-500 V
Max. charge / discharge current	50 A	
Max. charge / discharge power	10-15 kW / 10-11.3 kW	15 kW / 11.3 kW
Max. discharge depth	90%	

AC Grid

Grid voltage	3/N/PE 230/400V AC
Grid frequency	50Hz
Nominal power	10 kW
Max. active power $P_{E_{max}}$	11 kW
Max. apparent power	11 kVA
Nominal current	3 x 14.5 A
Max. current	3 x 25 A
THDI	3%
Power factor (cosφ)	1 (adjustable 0.8 leading - 0.8 lagging)

Backup power

Nominal power	10 kVA
Max. power (short time)	12 kVA (5 min), 15 kVA (10 s)
Switching time	10 ms from active operation, 60 s from standby

Inverter Efficiency		
Max. efficiency		98.4%
European efficiency	97.9%	
Max. efficiency for charging/discharging		98%
Safety and protection features		
DC-switch		Yes
PV reverse polarity protection	Yes	
Battery reverse polarity protection		Yes
Output short circuit protection	Yes	
Output overcurrent protection		Yes
Output overvoltage protection	Yes	
Isolation fault detection		Yes
GFCI	Yes	
Anti islanding		Yes
Internal PE-N bridge relay (Offgrid / EPS)	Yes	
Overvoltage protection		DC Type II, AC Type II
General Data		
Ambient temperature discharge/charge		-20° ... +55°C / 0°C ... +50°C
Air humidity rel.	max. 95% (non condensing)	
Max. altitude		4000 m (power derating > 2000 m)
Topology	Transformerless	
Parallel operation		Yes, max. 3 units
Mounting	On ground, secured to wall	
Ingress protection		IP65
Dimensions (W*H*D)	780 x 1760 x 240 mm (6 Bat.)	780 x 1620 x 480 mm (10 Bat.)
Weight	215 kg (6 Bat.)	315 kg (10 Bat.)
Cooling & noise	passive, <30dB @ 1m	
Communication interfaces		WiFi/LAN/Bluetooth (Monitoring App), RS485 (Smart Meter, HEMS), CAN (Battery), I/O inputs for RSE/DRM
Compatible HEMS	Solar Manager www.solarmanager.ch	
Display, UI		Status-LED-Panel, Monitoring App
Certification	Unit & Network protection i.a.w. VDE-AR-N 4105, VDE-AR-E 2510, EN 13849/60529/61000/62109/62477/62619, CE, RoHS In preparation: EN 50549, TOR	
Warranty & Battery Life		
Warranty Inverter & Battery*		12 years
Battery life**		10 000 cycles

* Guaranteed battery capacity during warranty period ≥ 65% of nominal capacity

**Discharge rate ≤ 0.2 C, DoD ≤90%, Cell temperature range: 22°C-28°C, remaining capacity @ EoL ≥ 65%

Subject to changes and errors, all information without guaranty

ALFRED All-in-One PV energy storage system

Hybrid-inverter, 2 MPPTs, high input current for latest generation of PV modules

Modular LFP-battery, capacity 9.6 - 24 kWh

True 3-phase backup power system with full inverter nominal power, black start capability

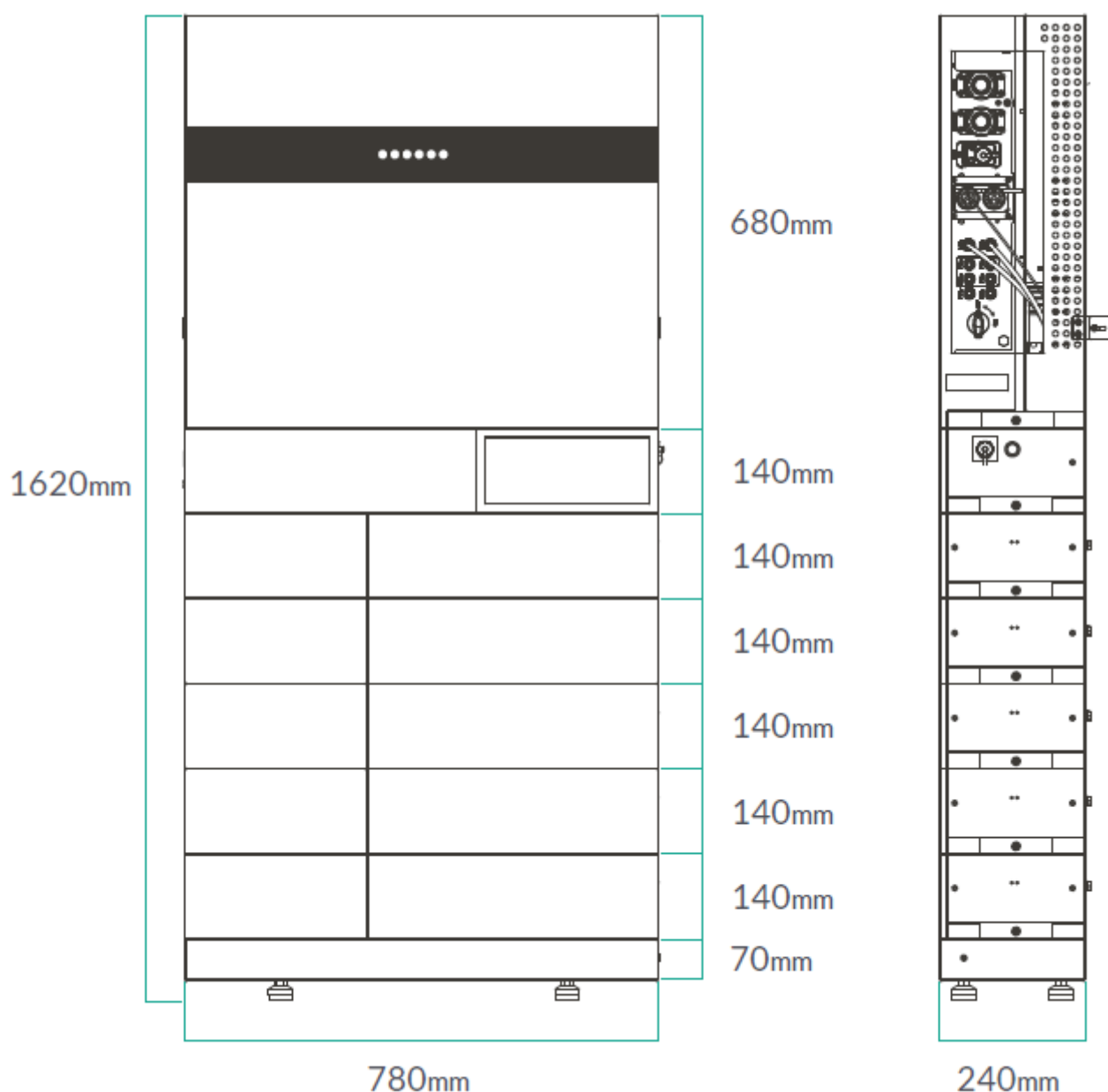
Ingress protection IP65, outdoor installation

Quick installation with minimized wiring work

Easy commissioning

Monitoring and settings via App, either local (bluetooth) oder remote (WiFi/Ethernet)

Award-winning clean design



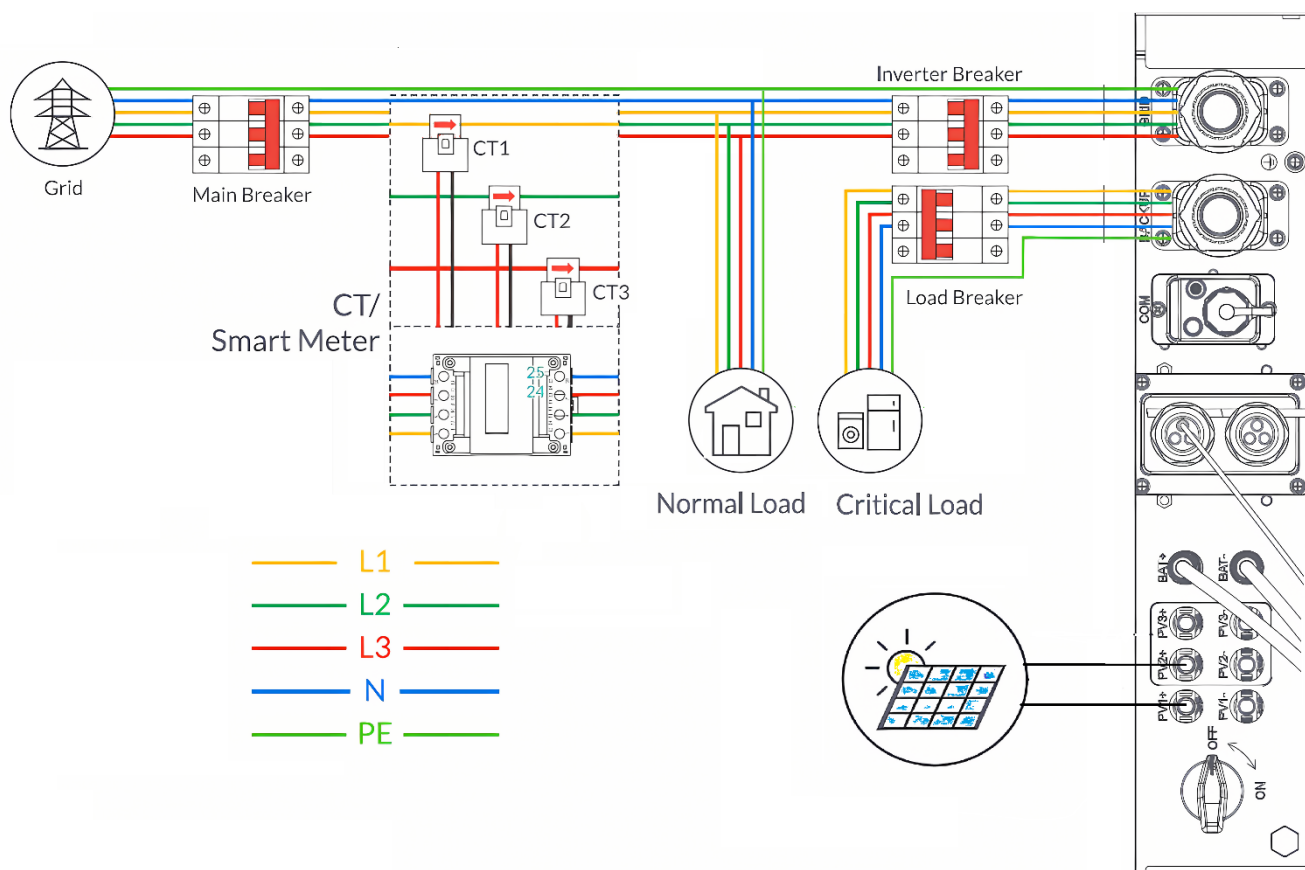
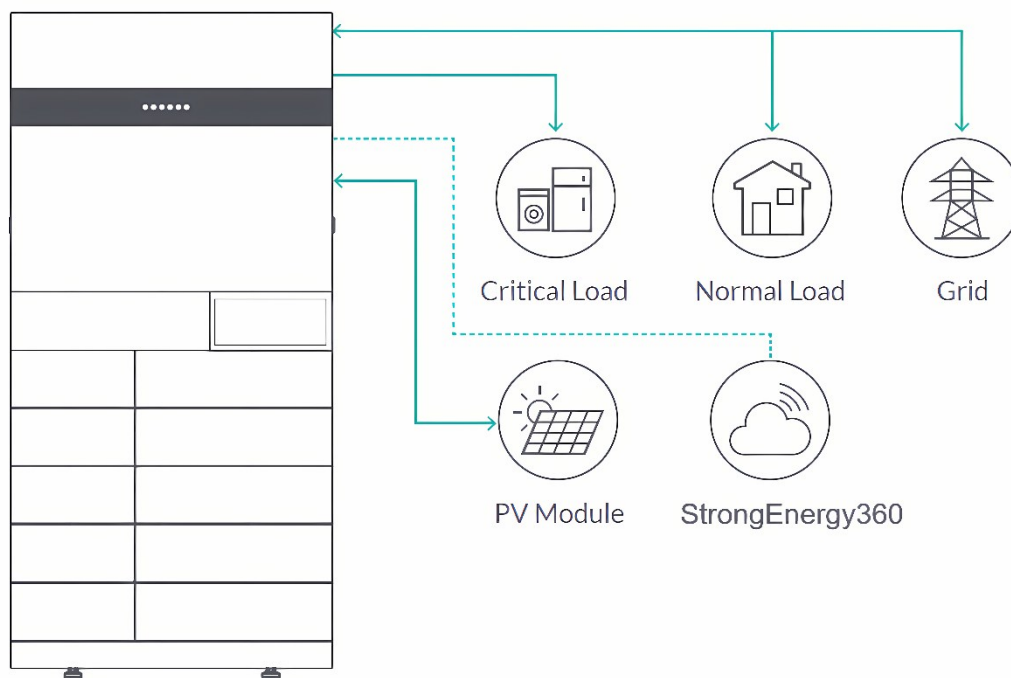
Standard-Installation

Backup-power connection on inverter, use optional

- Integrated grid disconnection relay

Power sensor on grid connection point:

- Current transformers connected directly to CT input on inverter
- Optional: Smart Meter DTSU666 with RS485 modbus data connection



Switchbox (optional)

Pre-wired distribution box

- Blends smoothly with ALFRED overall design, adds 240 mm to system height
- Overall height for ALFRED with 6 battery modules + Switchbox: 2000 mm
- Pre-wired connectors for fast & easy hook up to inverter

Integrated 80A Smartmeter

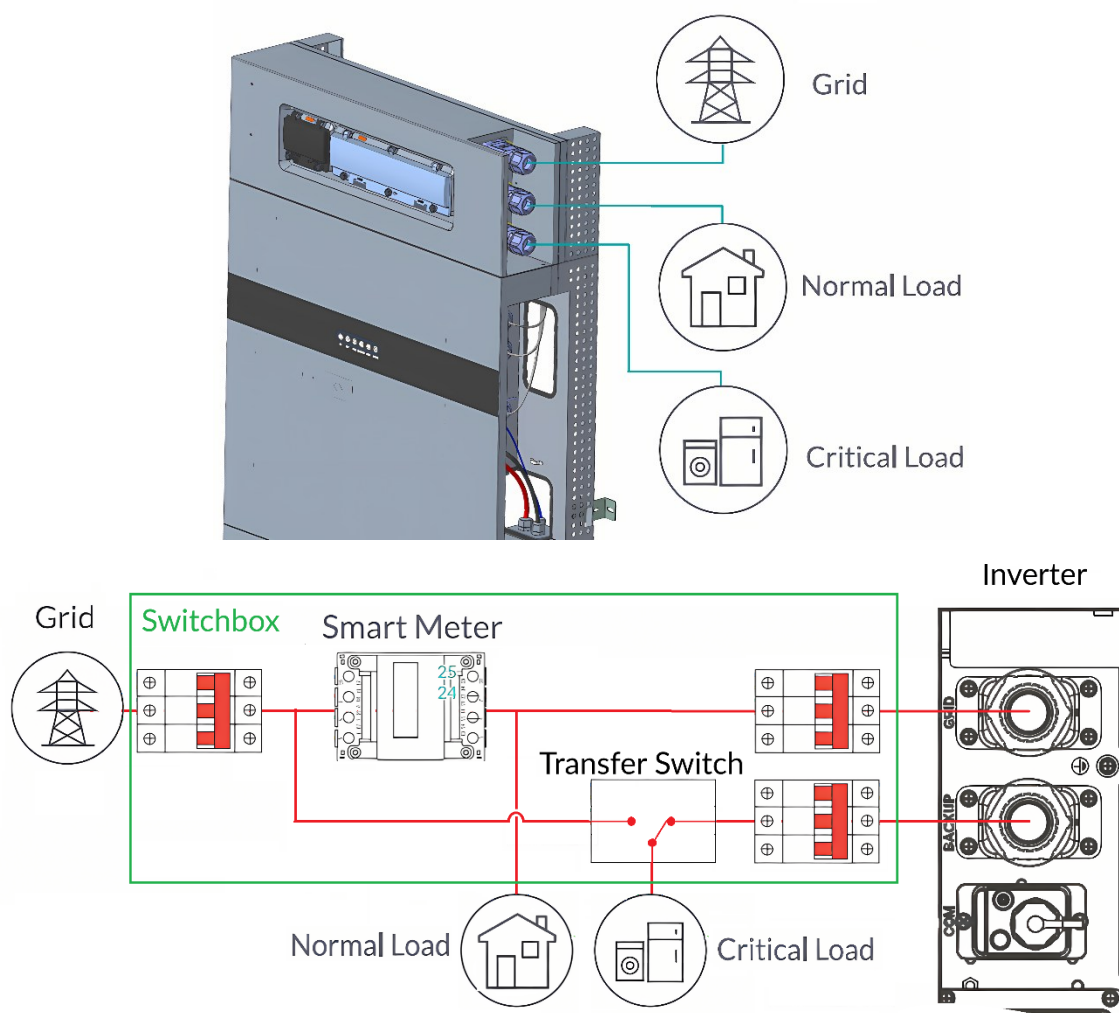
- RS485 data cable prewired, just plug into RJ45 receptable on inverter
- No more mixed up phases or inverted CTs

Integrated automatic transfer switch for critical loads

- In case of grid failure, critical loads are powered from PV and battery
- In case of inverter shutdown, critical loads are automatically switched over to grid power

Minimal changes to main electrical cabinet required

- 3 x 32 A circuit breakers for grid, inverter, and backup integrated into Switchbox
- Just run 3 x 5-wire cables from main cabinet to Switchbox
- No mounting space for additional circuit breakers or Smartmeter needed in main cabinet



*Switchbox makes the installation easier -
and the power supply more reliable!*